

Problem

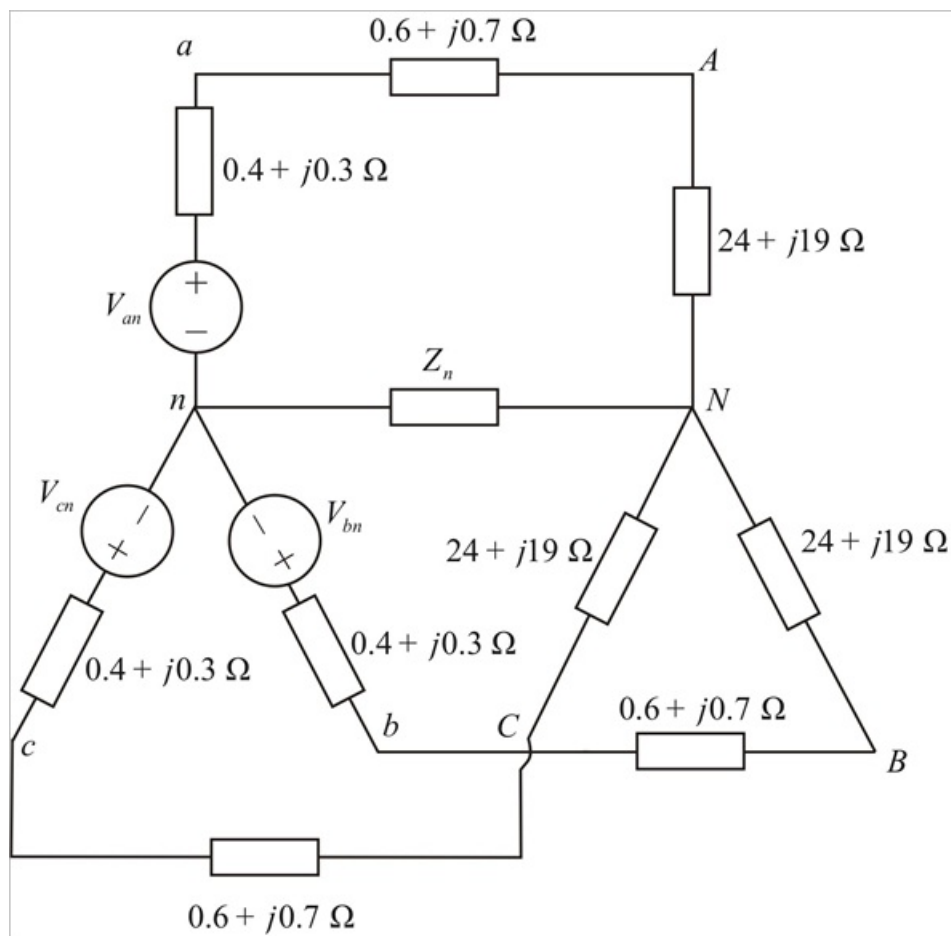
A Y-connected balanced three-phase generator with an impedance of $0.4 + j0.3 \Omega$ per phase is connected to a Y-connected balanced load with an impedance of $24 + j19 \Omega$ per phase. The line joining the generator and the load has an impedance of $0.6 + j0.7 \Omega$ per phase. Assuming a positive sequence for the source voltages and that

$V_{an} = 120\angle 30^\circ \text{ V}$, find: (a) the line voltages, (b) the line currents.

Step-by-step solution

Step 1 of 5

According to given values the $Y-Y$ balanced system is



Step 2 of 5

Given that $V_{an} = 120\angle 30^\circ \text{ V}$

Therefore

$$V_{bn} = 120\angle 30^\circ - 120$$

$$V_{bn} = 120\angle -90^\circ \text{ V}$$

$$V_{cn} = 120\angle 30^\circ - 240$$

$$V_{cn} = 120\angle -210^\circ \text{ V}$$

(a) \

The line voltages are

$$V_{ab} = \sqrt{3}V_p \angle 30^\circ$$

Where

$$V_p = 120^\circ \text{ V} \quad \boxed{\because V_{an} = V_p \angle 0^\circ}$$

$$V_{ab} = \sqrt{3} \times 120 \angle 30^\circ \times 1 \angle 30^\circ$$

$$\therefore \boxed{V_{ab} = 207.85 \angle 60^\circ \text{ V}}$$

$$V_{bc} = V_{ab} \angle -120^\circ$$

$$V_{bc} = 207.85 \angle 60^\circ \times 1 \angle -120^\circ$$

$$\therefore \boxed{V_{bc} = 207.85 \angle -60^\circ \text{ V}}$$

Step 4 of 5

$$V_{ca} = V_{ab} \angle 120^\circ$$

$$V_{ca} = 207.85 \angle 60^\circ \times 1 \angle 120^\circ$$

$$V_{ca} = 207.85 \angle 180^\circ \text{ V (or)}$$

$$V_{ca} = 207.85 \angle 180^\circ - 360^\circ$$

$$\therefore \boxed{V_{ca} = 207.85 \angle -180^\circ \text{ V}}$$

Step 5 of 5

(b)

The line currents are

$$I_a = \frac{V_{an}}{Z_Y}$$

Where $Z_Y = 0.4 + j0.3 + 24 + j19 + 0.6 + j0.7$

$$Z_Y = 25 + j20$$

$$Z_Y = 32 \angle 38.66^\circ \Omega$$

Hence

$$I_a = \frac{120 \angle 30^\circ}{32 \angle 38.66^\circ}$$

$$\therefore \boxed{I_a = 3.75 \angle -8.66^\circ \text{ A}}$$

$$I_b = I_a \angle -120^\circ$$

$$I_b = 3.75 \angle -8.66^\circ \times 1 \angle -120^\circ$$

$$\therefore \boxed{I_b = 3.75 \angle -128.66^\circ \text{ A}}$$

$$I_c = I_a \angle 120^\circ$$

$$I_c = 3.75 \angle -8.66^\circ \times 1 \angle 120^\circ$$

$$\therefore \boxed{I_c = 3.75 \angle 111.34^\circ \text{ A}}$$